GENETIC ALGORITHMS

Representing N-queens problem in memory can be performed through:

- Two-dimensional array
- Bit-string

Example:

Represent the following 5-queens problem using

a) 2D-array representation **b) bit-string representation**

				*
*				
		*		
	*		*	

Solution:

2D-array representation

0	0	0	0	1
1	0	0	0	0
0	0	1	0	0
0	1	0	1	0
0	0	0	0	0

bit-string representation

42325

Question:

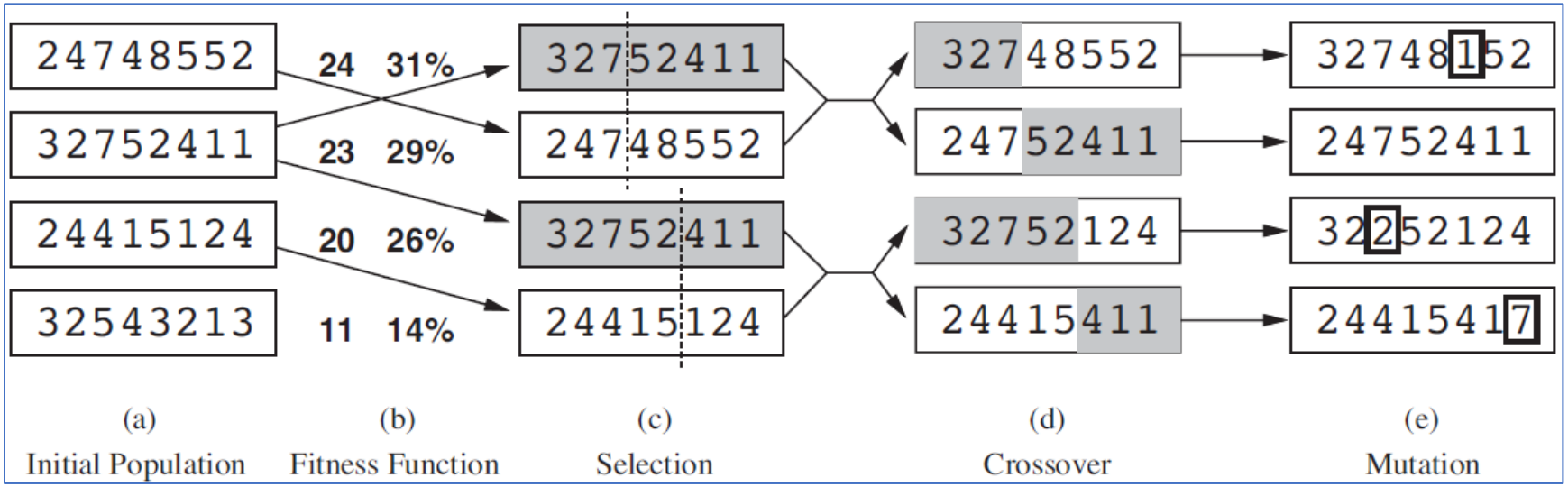
In case of N-queens problem, what is the value of h1(n) and h2(n) that indicate the solution has been reached?

Genetic Algorithms

The main concepts of GAs

Steps of GAs

- Generate a population
- Calculate fitness value
- Find the probability
- Selection
- Crossover
- Mutation


24748552
32752411
24415124
32543213
24 31%
23 29%
20 26%
11 14%
32752411
24748552
32752411
24415124
32748552
24752411
32752124
24415411
32748152
24752411
32252124
24415417
(a)
Initial Population
(b)
Fitness Function
(c)
Selection
(d)
Crossover
(e)
Mutation

